ActInf Textbook Group ~ Cohort 2 ~ Meeting 14 (Chapter 6, part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_2 | Cohort 2 ~ Meeting 14 (Chapter 6, part 1) | 55:18

Hello, welcome, thanks everyone. It's February 1st, 2023. We're in Meeting 14, Cohort 2 of the Paradol textbook. We're on Chapter 6. So, today we can look over any and all of the questions on Chapter 6 that have been raised. We can also go to the text. But just to begin, does anyone want to bring up any quote or topic or anything about Chapter 6 or any section from the text or question that someone wants to go to first or a general point about Chapter 6?
Ali, and then anyone else?
Well, I think now that the white paper from Versus Lab is published, it might be a good idea to read Designing Ecosystems white paper along with Chapter 6 because it can provide a much wider perspective and see how the research in developing these kinds of active inference based AI is currently developing and to see how the plan for the future developments have already been discussed, especially in the case for sympathetic and shared intelligence, which is kind of long-term plan to develop those kinds of intelligence systems. So, yeah, I think this paper can immensely help to see the big picture related for designing these kinds of active inference based models.
Awesome. Thanks. Yeah, the first 20 pages of the paper have great points, but just to jump to the timeline part. It's 2023. This was released at the end of 2022. And in the coming years, here are some doubling timelines that we might expect or prefer. So that does help set the stage for Chapter 6.
Thanks. Anyone else? Just unmute or just raise your hand. Chapter 6 is a recipe for building active inference models. So it's not the fast food restaurant yet, but it's a recipe for the home and potentially industrial chef. It covers the essential steps to design an effective model. And that is going to be taken in like a staged way. And starts on page 105 in the textbook. So shall we look at the questions?
Does anyone have a specific question they would like to jump to or any general points?
does anyone have a specific question?
Otherwise, let's look at the recipe and then see if there's things that we can add. And then in this active inference model recipe, let's see if we can flesh it out with what we're seeing and learning from doing modeling.
And also how it's similar and different from other modeling recipes.
How different is this than doing a linear regression model? How different is it than doing some other type of modeling or analysis?
So give me six chapters to model active inference and I'll spend the first four on the generative model.
So,
Carl Lincoln.
Which system are we modeling?
What is the appropriate form for the generative model?
How to set up a generative model?

How to set up the generative process?
So those have been copied out here.
Does anyone just want to give some comments on these stages or how would one just briefly summarize
what each of these four stages are and what they do?
Or just any feature about these four steps?
I guess one thing is they need to be discrete these steps and that's already a decision
that it's we made.
Do you mean the steps are separate from each other?
Yeah.
And we need to have categories that's if we have these hidden states in the Markov model,
these are separate states.
And I guess that's a kind of a problem and a restriction also.
Are there other modeling approaches that you think don't have such restriction?
Yeah.
For instance, in neural networks, these transformer models where you can type in a whole picture,
for instance, or something like this?
I may be wrong.
So maybe there's categories too.
Yes.
Well, maybe a related question is when we specify what is a hidden state and what is an observation
in our generative model?
Are we kind of locked into that?
Or is it possible for something like a pre-POMDP that describes structure?
So potentially this is between steps two and three.
The hidden state is going to be an image and then it gets set up with a certain dimensionality.
So, anyone can raise their hand.
Let's just look through the questions, see what we can bring into the modeling page and
which ones are remaining to be needed.
Why is modeling of the generative process a necessary step for building an active inference
model?
Okay.
We have a lot of notes on here.
What would somebody just give on a first pass?
Why is the generative process a necessary step for building an active inference model?
I guess on a very high level, because the generative model is ultimately creating a representation
of the generative process.
If we want to create that model, we have to allow for the generative model to be able to
model the generative process.

So, we also need to model the generative process.
Yeah.
One aspect there is the generative model is, or the generative process is providing the observations.
And then the sort of second layer on that is the generative model is working with, or is building representations, or is a good regulator of, or is needing to maintain the requisite diversity to deal with the generative process.
So, it would be like asking, why is modeling of the grid important in a NetLogo agent-based simulation?
It's like, well, you got two kinds of things.
The agent's moving around the grid, and you have the grid.
So, in this active inference particular partition, we have the generative model.
That's the active entity.
And then we have the generative process, which could be another active entity, or it can have any structure.
So, like in model stream 7.2, the generative model was an active agent.
The generative process was just an if-then logic.
Ali?
I also think that these distinctive steps that have been outlined here in this chapter as a recipe to designing any active inference model are not necessarily, are not specifically distinct or even sequential steps.
Because, for example, in the first step, it says, which system are we modeling?
But on the other hand, the choice of, sorry, the choice of the system that we're modeling would inevitably have an effect on how, on the type of the generative model or the type of the generative process we're dealing with.
So, in a sense, all of these steps are kind of intertwined into each other, and they're not necessarily have to be seen as a kind of in a sequential order.
But rather, in a kind of, from a hierarchical point of view, we just need to have which component of the modeling process are we dealing with.
So, for example, we can possibly begin by identifying the generative process and then look at how we can optimally represent the generative process by a suitable generative model.
So, what I'm saying is, these four steps are not necessarily something that, to be taken on the face level.
They are just laying out the territorial map of modeling those systems.
And we might begin from each one of them and then proceed to any other, depending on the situation we're involved in.
Yeah.
Thank you.
So, it may be seen as an atemporal partition, not necessarily the order that a traditional

recipe would have.
Let's, let's continue on.
And so we can just touch on each of the questions.
Okay.
What are the four steps?
Some of this material we moved over to this distinct model recipe page, just so that we can develop that in its own section.
Okay.
What are the four steps that are being addressed on that page?
In figure 6.1, what do the unidirectional and bidirectional arrows mean?
What's the thought anyone has on this?
Okay.
So, in a Bayesian graph, the nodes represent random variables and the edges represent statistical influences.
Those can be represented in an undirected fashion if there's no arrowheads.
And that's a lot like a structural equation model.
Like if we were dealing with linear notions of correlation, Pearson correlation, we could make a structural equation model.
And then the edges would be like correlations between two variables, which are undirected, of course.
So, in a Bayesian causality framework, like Judea Pearl et al., it is possible to have a bidirectional arrow, which is a lot like an undirected arrow.
But also you can have a unidirectional arrow such that like changes in one variable cause changes in another, but not asymmetrically, especially if you consider time series information.
So, in the work of Aguilera et al., how particular is the free energy, how particular work?
I think there are several papers and a stream on it.
So, they analyze which topologies of the action perception loop have which properties.
So, one can imagine like a simple cybernetic action perception loop, like which we could just call like around the clock.
So, no line in the middle, just internal states only influence action, no backwards arrow.
Action only influences internal, internal on sense, sense on internal, and all the self loops.
That's like a simple around the clock topology of the action perception loop.
So, then we can ask, alright, so from that around the clock causal framing, what extra edges and arrowheads exist in the way that it's shown in figure 6.1?
There are three.
There's the connection between active and sensory states.
So, the blanket states have an additional relationship.

And there's this somewhat interesting symmetry where active states have an arrow back to internal states and sensory states have an arrow back to external states.

So, frankly, I am not sure to what extent people are just showing representations of the way that previous models have shown it.

So, the other thing is, I think, is that the other thing that we can see is that the other could be a fully connected matrix with a value in every cell.

Or, it could be like just the identity matrix would be like four states that are independently evolving and not influencing each other.

So, different topologies of this particular partition are going to have different properties.

And I think the most general approach would be to do structure learning and Bayesian model selection on different topologies for the action perception loop.

So, what do they mean?

Well, they mean that you are comparing models with the constraint or the availability for that parameter value to be non-zero.

If you delete the arrowhead of a certain kind, then you're basically testing models within a restricted class where that parameter has been fixed at zero.

So, if you want to fix the effect of sensory states on external states to zero, maybe you think that's, quote, unrealistic.

So, you want to fix that to zero so that you have increased statistical power to resolve other parameters.

That's valid.

Although, one of the most intuitive reasons to do so, which is like, but sensory states don't influence external states in the real world.

It's like, yes, but that's the map territory fallacy.

This isn't the causal structure of reality.

Ali, or anyone else?

Actually, it might be helpful to look at figure two from the paper, Path Integrals, Particular Kinds and Strange Things,

and especially the different typologies of the particular kinds provided there, being inert particles, active particles, conservative and strange particles.

So, briefly, they define inert particles as the kind of particles or systems with no active states.

And in the order of increasing complexity, one step further or one level up would be the active particles, which is a particle with a non-empty set of active states.

But then, ultimately, we, and also the conservative particle is an active particle whose particular states follow paths of least action.

But ultimately, we reach a kind of definition, I believe, for the first time, for strange particles, which are defined as conservative particles whose active states do not directly influence internal states.

So, this paper claims that only this fourth type of particles, namely the strange particles, have the capacity to be a kind of sentient agent or any kind of intelligent agent, depending on how we define intelligence or sentience.

But in any case, all of these kind of this strange particles, somehow subsumes all the other kinds of particles, but with varying levels of complexity assigned to each of them.

Thank you.

Yes.

Good connection here.

We see, so interestingly, we don't see the backwards arrow.

Here, we see the around the clock and the intra-blanket connections.

But we don't have the backwards edge.

Yeah, that's because these kinds of strange particles

have not been developed into fully sentient agents yet.

So, to be fully sentient is to be able to actively influence all the active and sensory states.

So, I believe that's why we don't have those bidirectionality in this picture for strange particles yet.

Thank you.

Yes, it's almost like this is one circuit diagram.

It's like we have four CPUs on our circuit board.

That's the particular partition.

And then how we wire up the topology of the particular partition is like the hardware of our circuit board.

And then what the dynamics are is like the software of our circuit board.

Once we start mixing in these taxonomies of particles of increasing sentience and cognitive complexity,

and then we start mixing in the blanket index like it's a big space, but also it's a very semantic space.

And one advantage of FEP and active modeling is we can develop special or reduced cases of particular things that are subcognitive,

but they're still broadly considered within a cognitivist framework.

Whereas something like thousand brains, it may be a super effective architecture for inference or generalization or any other number of advanced cognitive tasks.

However, it cannot be constrained or projected down to the case of like a ball rolling down a slope.

Whereas Bayesian mechanics, Bayesian physics, and this taxonomy of particles helps us bridge the continuum with expressive modeling framework that spans the simplest, least active particles to open-ended complexity and intelligence.

Michael?

Michael?

Michael?

Michael?

Michael?

Michael?

Michael?

Michael?

going to the brain so that the nerve ends in the spine somewhere and then this action

is produced by a place in the spine and then it doesn't go to...

But it depends on what you call an internal state, right?

So if the spine that triggers this reaction is not part of an internal state...

Well, I'm a little bit lost in understanding how we can understand that.

Yeah, thank you for the comments.

So super important point, like it depends what is internal states.

It's not like states even within a fixed...

These are the only variables on the table.

Whether one is internal or external is of course whether we're choosing to model it

as the generative model or a generative process, like which side of the blanket and there's a symmetry there.

And then it's not just like this variable is an internal state.

It's always like the fourfold particular partition comes into being in the same time.

Like the blanket states are those that make internal and external states conditionally independent.

So, but that blanket state with respect to X and Mu, it might be an internal state with respect to something else.

So it does definitely depend on it.

And I think now to the question of the connection.

Here's from Emperor's New Markov Blankets by Brunberg.

I think the edge may have to do with just the...

The fact that there is an arrowhead doesn't mean that the parameter value is non-zero.

It just means that we're describing families of models where there can be a non-zero influence.

We're not a priori fixing it to zero.

But you could conceivably remove this edge and just fit the simple around the clock or around the clock with the two backwards edges.

So, because we're making models of maps, not territories, we don't need to be constrained to a physical anatomical realism with respect to what these edges are.

And so it's easy to sort of contrast the plausibility of these edges with anatomical realism.

But these are not anatomical edges.

So, it's just opening up the family of models under consideration to a non-zero statistical influence between these states.

And it also may inherit from the question of co-parents and also the developments of what people call a frist and blanket.

Whereas the Markov blanket formalism by Perl was initially characterized on an undirected graph and there was only one kind of blanket state.

It was just the set of nodes that make internal and external states conditionally independent.

And you could imagine a sort of clean frist and blanket with like a clean division of labor where there's some of the nodes of that blanket set are strictly outgoing.

And some of the nodes are strictly incoming.

So, that could happen.

You could dictate design systems that have that characteristic.

And that may map onto anatomical realism.

But also, you might want to consider parameterizations of models that don't have such a clean partitioning.

Like there's a dual functional node that has an incoming and an outgoing statistical dependency.

And that doesn't have to map onto any actual sensor or actuator in the world.

It's just a statistical node.

But isn't...

But isn't...

Okay, I can see that one can design such a system.

But does it make sense?

I mean, if there's an infinite loop between active states and sensory states that mutually manipulate themselves,

but they have neither an impact on an external world, not on an internal world, not on an internal world, how...

Of course, we can assume this, but what is the reasoning there?

I mean, does it make sense?

If something happens and we cannot see it, not outside, not inside, what is the point?

It's just like a full Turing machine that generates a derivation tree with huge number of internal nodes that just disappear.

And one has not a track of what happens there.

Yes.

Thank you for the comment.

So I think one way to think about this topology, this wiring of the particular partition.

Here's from that Path Integrals Particular Kind Strange Things paper.

This is how we're modeling flows over states in the FEP.

We have the flow over the total particular partition, X, is like a tuple.

It's a four-length vector over internal, external sense and action states.

And the flows of each of these nodes are a function of if it were all by all connected.

So if there was like an edge here, then you could just like the fully connected model would be...

Each of the flows would be a function of all four other variables.

And you can write that out if you wanted.

But importantly, in these bottom two, the action and the internal, or the autonomous states, don't have a direct reliance on external states.

The eta is not in the bottom two.

And then, importantly, the external and the sense states don't have a direct...

They're not taking as an argument internal states directly.

That is like the minimal cybernetic constraint.

We might be then interested in constraining further.

But...

And FEP may be general.

It may apply to the all-by-all case as well.

But this is the minimal constraint that allows autonomous states, active and internal, to be under perceptual and action selection imperatives.

So, in any given case, the topology itself could be a structure learning question.

Or, given a fixed topology, parameterizing it is going to be the question.

But does it make sense?

It's like saying, does a given structural equation model make sense?

It just is a function of what data you have.

Let's see if we can just touch upon subsequent questions.

Is there a common good representation or rubric for evaluating GMGP?

Does anyone have any thoughts?

Does anyone have any thoughts?

Does anyone have any thoughts?

Well, one approach that's not even ACTIMF specific is using the Bayesian information criterion.

Or its cousin, the AIC.

The Ikkie information criterion.

and the BIC and basically the AIC reward good fit and they penalize models with more parameters.

So the BIC is widely used to establish the Pareto optimal trade-off of model complexity

and model accuracy. So if one is interested in what is a good generative model, generative process

fit, if you just want the sheer fitness, you're going to end up on the slippery slope towards high

dimensional generative models. Because adding new parameters appropriately strictly increases model

likelihood. Because new parameters are always sucking up variance that was left unexplained

in the noise term by whatever you had previously. So we don't just want to simply increase the

likelihood blind to all other factors. We want some sort of balanced measure that penalizes adding

more parameters. So that's one approach for model evaluation in the Bayesian space. And another

measure that is commonly used is a Bayes factor, which is a relative measure of bit amongst multiple

Bayesian models. So the Bayes factor is a ratio between two Bayesian models. And so it can be used,

people have kind of heuristics, just like there are heuristics for p-values,

not even going into the whole question about p-values here. But there are also heuristics

for Bayes factors, whether the evidence of one model over another is like substantial, strong,

decisive, et cetera. And Bayes factor allow any models to be compared with each other. Whereas if we were in

the parametric modeling space, we would only be able to test using the hierarchical likelihood ratio test, the HLRT for parametric models. So for parametric models, you can only test nested models. Like if you're doing an ANOVA, you can test an effective A and effective B, and then you can compare that with whether there's an

interaction between A and B. But you would have a hard time testing a model with A and B versus a model with B and C.

Unless you put that into a nested parametric framework, Bayesian statistics gives us a lot more flexibility to compare models.

So model selection, evaluation, comparison, those are big areas. Short answer, we're using Bayesian statistics.

So we have the most modern tools for evaluating that. In the specifics, it's always going to be the specifics.

But in terms of the tools for model selection, Bayesian statistics has the best tools.

All right. How does the particular partition, so 6.1, which we were looking at earlier,

the sort of proto-particular partition, the dyadic partition here, informal, relate to the POMDP in figure 4.3?

What would anyone say?

These are two common figures that we see,

but I don't see X, mu, U, and Y here.

So interestingly, the X and the Y in the continuous time representation,

they actually do map on to the particular partition,

and kind of with a V.

whereas the discrete time POMDP

does not use this exact notation.

What would anyone say?

Why do we have this closed

cybernetic

partition,

but then it looks like we're moving into

a POMDP model

with a different topology?

I guess in the

POMDP model,

there should be also

an arrow going backwards

to the beginning of the...

So if we...

In a state on the right side,

there should be somehow

a loop back to the beginning,

which is your implicit somehow.

That is called sophisticated active inference,
where in the past, present, and future,
there can be an inference over the past, present, and future.
Okay, yeah.
So it is possible,
but it's not the base case.
Yeah.
Yeah.
No.
Yes.
So this is unrolled in time, no?
Actually.
Yes.
Exactly.
Exactly.
This has no time.
So this is unrolled in time.
The particular partitions
are not defined in time.
They're defined as functions of other states in the partition
because what a Bayes' causal graph represents
is not as a function of time.
So it has to be put into a form
that allows us to explicitly enumerate time.
And then in chapter four,
and then again, we'll see in chapter seven and eight,
we have two ways to deal with time broadly.
Discrete time, continuous time.
Discrete time actually models
discrete slices of time sequentially
and generates parameterizations
for like hidden states and observations
at different explicit time points.
In contrast,
the continuous time generative model,
which is being shown
like with these one, twos, and threes
instead of the D, B, and A,

they're being shown juxtaposed in figure four, three
to emphasize their structural similarities.
However,
whereas the discrete time model
is explicitly predicting states
and observations at different time points,
the continuous time model,
its B matrix equivalent
is not the next state,
but rather the derivative of the state.
And so the continuous time model
has more in common with a Taylor expansion
or a Volterra expansion
or generalized coordinates of motion.
So in either case,
POMDPs help us
focus our statistical
and informational power
on constrained sets of models
that are not too restrictive
if we're willing to abide
by certain
standard
modeling
practices
like that the past
can't influence the future
except through the present
or that like a derivative
can't have an influence
on its second derivative
without influencing
the first derivative first.
So under those
standard
modeling
assumptions
that

reduce the state space
vastly,
like one can imagine
if there was 10 time points
or 10 derivatives
and you did all by all,
you'd end up with a lot of edges.
But if you constrain it to
time points can only influence
next time points
and derivatives can only influence
the next derivatives,
then
your model scales linearly
with time points
or derivatives.
rather than
supra linearly.
So
POMDP
is one
implementation
architecture
that generates
generative models
which are compatible
with the continuous
flow
Bayesian physics
underlying.
Just like you might have
some flow
or field equations
for electromagnetics,
and then you have
some discrete time
realization
in your simulation,

that is the relationship.
I have a question.
So if you say it's like a Taylor series,
then this is somehow deterministic,
isn't it?
I mean,
in the beginning,
in the Taylor series,
you say what functions
you want to decompose,
and then
all the
parts are a little bit
deterministic.
I think,
but is it
what we are trying
to model here?
Or am I
completely wrong?
I mean,
a series,
you already,
I mean,
it's maybe infinite,
but
you know
from the beginning
what the
what the
individual
components are.
Anyone else
want to give a thought
on that,
or I can
add something?
Well,

it can be
deterministic.
So you could have
a model
where,
for example,
like
there isn't
like
there's
an identity matrix
for A.
So like hidden states
are fully observable.
Or you could have one
where the state transitions
are fully deterministic.
as well
in the discrete time case.
Okay.
Now in the
continuous case.
Is it
deterministic?
Well,
from a given
hidden state,
you could have
a deterministic
or non-deterministic
emission
of observations.
You can have
probabilistic
action selection
that influences
how states
change through

time.
And the way
that they
change through
time
could be
sampled,
hence
also not
deterministic.
take
and I think
the
Taylor series
and
variational
inference
have a lot
in common
in that
you do
have to
commit
to like
a given
family
of
distributions
that you're
wanting to
fit.
But then
within a
given
variational
family,
so you have
some data

set
and you
say,
and now I'm
going to
fit this
type of
distribution
to it,
you might
fit that
with the
stochastic
gradient
descent,
or there
might be a
deterministic
algorithm to
fit it.
but
it's a
little bit
too broad
of a
brush
to
think about
the bigger
picture of
using these
models and
simply say
that they're
deterministic
or
non-deterministic.
I hope that

makes sense
because it
is a very
subtle
question.
Yeah,
okay.
Yeah,
thanks.
I'm
wondering if
you could
have a
non-deterministic,
can a
function have
a stochastic
derivative?
Well,
I don't
know,
but I
mean,
somehow if
we think
of the
environment
that produces
new
input
and we
don't know
that
environment,
then somehow
in undetermination
factor is
there,

right?
Yes,
the generative
process is
not shown
here.
The generative
process,
external states,
is handing
observations in.
So the GM,
to kind of bring
back the
question,
it unrolls
and it
shows the
agent's
perspective.
this is
necessary and
sufficient to
define the
agent.
But here's
the great
unknown that's
handing the
observations to
the agent.
So the
GM can be
seen as like
the cognitive
architecture of
the agent.
it is not

describing the
total scenario
that the
particular
partition is
showing.
So that's
one other
difference.
This S
is not
the actual
external states.
This is the
agent's
variational
estimate of
external states.
Ali?
I think
the remark
on page
24 of
Path
Integral's
paper might
help to
understand this
issue a little
bit better,
especially the
second paragraph,
which says
that from the
perspective of
an observer,
behavior will
appear to be

fulfilling the
epistemic and
pragmatic
imperatives
afforded by
expected free
energy.
So it's
basically the
control versus
planning or
homeostasis
versus
allostasis
when we
look at
the inference
from the
point of view
of the
internal states
of the
agent.
Because only
from that
perspective can
we talk about
reducing the
generalized free
energy,
right?
So there's
a fundamental
difference between
looking at this
agent from
its internal
state's point

of view and
looking at it
from the
external,
from the
point of view
of an
external observer.
I don't know
if it makes
sense, but
although they're
complementary to
each other, but
in a sense,
they're fundamentally
different.
Thank you.
And just to
look ahead to
nine, though,
I wonder if
this might
be useful
to see in
six, and
it relates to
the map
territory
question.
So here's
figure 4.3 is
in the middle.
We're
parameterizing a
POMDP with
real computers
about a

rat in a
T-maze, or
an ant in a
colony, or a
cyber-physical
entity.
So we're
parameterizing
this POMDP,
but that
POMDP is
in a
parameterized
environment as
well.
And six
kind of
brings us
to that,
but it
would be
awesome to
see more
connective
tissue
between
the
particular
partition,
which is
where the
physics
is
inheriting
from,
flows on
the
particular

partition,
and how
that relates
to this
view from
the inside
of a
cognitive
agent.
Okay,
in the
final
minutes
of this,
chapter six.
Okay,
last week,
I think we
explored a
little bit
about how
B and
A are
two different
variables.
They're not
directly connected
to each other,
but they kind
of work as
a team.
Like B
increments the
hidden state
along, and
then the
hidden state
emits an

observation.
Good
regulator,
we talked a
little bit
with Bronwyn
at office
hours.
We explored
good regulator
theorem and
cybernetics,
requisite
diversity.
We talked
about what
happens to
bad
regulators.
We explored
also in the
office hours
last week a
little bit
of the
discussion
around the
blanket trick
and about
how affordance
is handled
in Act-Inf
and some
similarities and
contrasts with
how it is
handled in
other areas

of like
other threads
of thinking
in ecological
psychology.
In the
coming
second
discussion on
chapter six,
we can
return to
these bottom
three
questions.
temporal
depth,
cognitive
systems
outside the
brain,
and
similarities
and
differences
with active
inference and
other modeling.
Any final
remarks on
this?
all right,
awesome.
I will stop
recording.
Let us take
a two-minute
break and

then we
will begin
for cohort
three.
Thank you
all.
Thank you
all.
Thank you
all.
Thank you

Thank you

Thank you